

TIC TAC TOE IMPLEMENTATION

```
board = [  
    ['O', ' ', 'X'],  
    ['X', ' ', ' '],  
    ['X', 'O', 'O']  
]  
  
path_cost = 2  
  
print("Initial State:")  
for row in board:  
    print(row[0], "|", row[1], "|", row[2])  
  
for i in range(3):  
    for j in range(3):  
  
        if board[i][j] == ' ':  
  
            board[i][j] = 'X'  
  
    print("\nNew State:")  
    for row in board:  
        print(row[0], "|", row[1], "|", row[2])  
  
    print("Path Cost =", path_cost)  
  
    if ((board[0][0] == board[0][1] == board[0][2] == 'X') or  
        (board[1][0] == board[1][1] == board[1][2] == 'X') or  
        (board[2][0] == board[2][1] == board[2][2] == 'X') or  
        (board[0][0] == board[1][0] == board[2][0] == 'X') or  
        (board[0][1] == board[1][1] == board[2][1] == 'X') or
```

```
(board[0][2] == board[1][2] == board[2][2] == 'X') or  
(board[0][0] == board[1][1] == board[2][2] == 'X') or  
(board[0][2] == board[1][1] == board[2][0] == 'X')):
```

```
print("X Wins!")
```

```
board[i][j] = ''
```

OUTPUT

```
... Initial State:
```

```
O |   | X  
X |   |  
X | O | O
```

```
New State:
```

```
O | X | X  
X |   |  
X | O | O
```

```
Path Cost = 2
```

```
New State:
```

```
O |   | X  
X | X |  
X | O | O
```

```
Path Cost = 2
```

```
X Wins!
```

```
New State:
```

```
O |   | X  
X |   | X  
X | O | O
```

```
Path Cost = 2
```

VACUUM CLEANER IMPLEMENTATION

```
import random
import time

# Initial environment
locations = {
    'locA': random.choice(['Clean', 'Dirty']),
    'locB': random.choice(['Clean', 'Dirty'])
}

current_location = random.choice(['locA', 'locB'])

print("--- Initial State ---")
print(f"Room A: {locations['locA']} | Room B: {locations['locB']}")
print(f"Vacuum starts in: {current_location}\n")

# 4-step cleaning loop
for step in range(1, 5):
    print(f"[Step {step}] Vacuum is in {current_location}")

    if locations[current_location] == 'Dirty':
        print(f"Found dirt! Cleaning {current_location}...")
        locations[current_location] = 'Clean'
    else:
        print(f"{current_location} is already clean.")

    if current_location == 'locA':
        print("Moving Right to locB...")
        current_location = 'locB'
    else:
        print("Moving Left to locA...")
```

```
current_location = 'locA'

print(f"Current Status - Room A: {locations['locA']} | Room B: {locations['locB']}\n")

time.sleep(1)

print("--- Final Result ---")

print(f"Room A: {locations['locA']} | Room B: {locations['locB']}")

print("Cleaning complete!")
```

OUTPUT

```
... --- Initial State ---
Room A: Clean | Room B: Dirty
Vacuum starts in: locA

[Step 1] Vacuum is in locA
locA is already clean.
Moving Right to locB...
Current Status - Room A: Clean | Room B: Dirty

[Step 2] Vacuum is in locB
Found dirt! Cleaning locB...
Current Status - Room A: Clean | Room B: Clean

[Step 3] Vacuum is in locB
locB is already clean.
Moving Left to locA...
Current Status - Room A: Clean | Room B: Clean

[Step 4] Vacuum is in locA
locA is already clean.
Moving Right to locB...
Current Status - Room A: Clean | Room B: Clean

--- Final Result ---
Room A: Clean | Room B: Clean
Cleaning complete!
```